Example 6.9

The mean and standard deviation from 18 observations are 14 and 12 respectively. If an additional observation 8 is to be included, find the corrected mean and standard deviation.

Solution:

The sum of the 18 observations is = $n \times \bar{x} = 18 \times 14 = 252$.

The sum of the squares of these 18 observations

$$\sigma^2 = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2$$

$$12^2 = \frac{\sum x^2}{18} - 14^2$$

$$144+196 = \frac{\sum x^2}{18}$$

$$\frac{\sum x^2}{18} = 340$$

$$\sum x^2 = 340 \times 18 = 6120$$

When the additional observation 8 is included, then n=19 ,

$\Sigma x = 252+8 = 260$

Therefore, Corrected Mean = 260/19 = 13.68

$$\text{Corrected } \sum x^2 = \sum x^2 + 8^2$$

$$= 6120+64$$

$$= 6184$$

$$\text{Corrected Variance } \sigma^2 = \frac{6184}{19} - 13.68^2$$

$$= 325.47 - 187.14$$

$$= 138.33;$$

$$\text{Corrected Standard deviation } \sigma = \sqrt{138.33}$$

$$\sigma = 11.76$$

Example 6.10

A study of 100 engineering companies gives the following information

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